MALAWI INSTITUTE OF EDUCATION

MINISTRY OF EDUCATION, SCIENCE AND TECHNOLOGY

JUNIOR SECONDARY MATHEMATICS - FORM 1

Student Curriculum Textbook

Approved for secondary schools across Malawi.
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UNIT 1: NUMBER BASES & ARITHMETIC

1.1 Base Ten and Base Two Systems

The decimal system uses ten symbols (0 to 9) with base 10.
In digital computing and binary representation, base 2 is utilized.

Worked Example 1.1

Convert binary 1101_2 into decimal notation:
1*(2^3) + 1*(2^2) + 0*(2^1) + 1*(2^0) = 8 + 4 + 0 + 1 = 13.